INCH-POUND

MIL-R-87017B(CG)

20 March 1991

SUPERSEDING
MIL-R-87017A(CG)
2 December 1983

MILITARY SPECIFICATION

*RIBBON, DECORATIVE-TRIMMING, POLYESTER, GROSGRAIN (CG)

This specification is approved for use by the United States Coast Guard, Department of Transportation, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a decorative-trimming grosgrain polyester ribbon.

1.2 Classification. The ribbon shall be of the following types and classes:

Type I	- 5/16 inch wide (7.93mm)
Class 1	- Blue
Type II	- 1-1/2 inch wide (38.10mm)
Class 1	- Blue
Class 2	- Red

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, P.O. Box 59, Natick, MA 01760-0001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A *FSC 8315

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* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-P-636 - Boxes, Shipping, Fiberboard

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
* MIL-STD-2073-1 - DOD Materials Procedures for Development and Application of Packaging Requirements
* MIL-STD-2073-2 - Packaging Requirement Codes

* (Unless otherwise indicated, copies of federal and military specification, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. POSTAL MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

* 2.2 Non-government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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*TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Method No. 8-1988 - Colorfastness to Crocking: AATCC Crockmeter Method

(Application for copies should be addressed to AATCC National Headquarters, P.O. Box 12215, Research Triangle Park, NC 27709-9989.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606-5808.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard sample. The finished ribbon shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3). The standard sample for the Blue 3366 is identified under piece number 85049. The standard sample for the Red 3805 is identified under piece number 85048.

3.2 First article. When specified, the contractor shall furnish sample ribbon for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

3.3.1 Fibers. The fiber shall be 100 percent polyethylene glycol terephthalate when tested as specified in 4.5.

3.3.2 Denier. The warp yarns shall be 165 denier texturized polyester and the filling yarns shall be 150 denier texturized polyester. Testing shall be as specified in 4.5.

3.4 Physical requirements. The finished ribbon shall conform to the requirements specified in Table I when tested as specified in 4.5.

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Table I - Physical requirements

Characteristic	Type I	Type II
Width - inches (mm)	5/16 ± 1/16 (7.93 ± 1.58mm)	1-1/2 ± 1/16 (38.10 ± 1.58mm)
Weight - lbs/1000 yds - minimum (kg/1000 yds)	3.5 (1.74)	16 (7.94)
Warp ends - minimum		
Ground	38	222
Binder	4	8
Selvage	24	24
Filling yarns per inch (2.54cm) - minimum	50	50
Shrinkage % - maximum		
Warp	2	2
Filling	2	2

3.5 Color. The color of the finished ribbon shall be Blue 3366 for class 1 and Red 3805 for class 2 and shall match the standard sample.

* 3.5.1 Matching. The color and appearance of the dyed and finished ribbon shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7500K (±200K) with illumination of 100 (±20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300K (±200K) (see 4.4.2.4)

* 3.5.2 Colorfastness. The finished ribbon shall show colorfastness to laundering (3 cycles), perspiration, light (40 SFH), and crocking equal to or better than the standard sample when tested as specified in 4.5. As a limit of acceptability, or when no standard sample is available, the finished cloth shall show a minimum of "good" colorfastness to laundering (3 cycles) and perspiration and a minimum of "good" to light (40 SFH), and shall show a dry crocking rating not lower than 4.0 and a wet crocking rating not lower than 3.5.

3.6 Weave. The weave shall be a plain grosgrain with two picks weaving as one when tested as specified in 4.5.

3.7 Length and put-up. The ribbon shall be put up on rolls of 50 or 100 continuous yards (± 1/2 yard) (45 or 90 meter ± 45 cm)(see 6.2).

3.8 Fiber identification. Each roll of ribbon shall be labeled, or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

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3.9 Workmanship. The finished ribbon shall conform to the quality established by this specification. The occurrance of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article of the finished ribbon submitted in accordance with 3.2, shall be visually inspected for appearance and color. The sample shall be tested for physical and chemical properties in accordance with the methods specified in 4.5.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.4.

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4.4.2.1 Yard-by-yard examination. The required yardage of each roll shall be inspected and visual defects classified as listed below. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit shall be one linear yard. The sample size shall be based on inspection level II. The lot size shall be expressed in units of one linear yard. The number of rolls from which the sample is to be selected shall be in accordance with Table II. An approximately equal number of yards shall be examined in each roll. The acceptable quality level (AQL) for this examination shall be 1.5 major defects and 4.0 total defects (major and minor combined) per 100 units.

Defects	Classification Major	Minor
Hole, cut or tear	X	
Smash	X	
Broken or missing yarn		
- two or more continuous regardless of length	X	
- single, 2 inches (5.08cm) or more missing	X	
- single, less than 2 inches (5.08cm) missing		X
Floats or skips		
- multiple, more than 1/2 inch (12.07mm) in combined warp and filling directions, or single more than 2 inches (5.08cm) in length	X	
- multiple, 1/2 inch (12.70mm) or less in combined warp and filling directions, or single less than 2 inches (5.08cm) in length		X
Knot - oversized		X
Heavy filling bar or heavy place, clearly visible*		X
Kinks - multiple, clearly visible*		X
Open place or crack		
- opening at widest point three or more yarn diameters or sequence of smaller openings that may result in a weak spot	X	
Spot, stain or streak		
- clearly visible* 1/2 inch (12.70mm) or more in combined warp and filling directions	X	
- clearly visible* but less than 1/2 inch (12.70mm) in combined warp and filling directions		X
Crease or wrinkle embedded	X	
Filling yarn showing between ground and binder		X

* Clearly visible at normal inspection distance (approximately 3 feet)(91.4cm)

4.4.2.2 Overall examination. Each defect listed shall be counted not more than once in each roll examined. The sample unit shall be one roll. The number of rolls selected for yard by yard examination shall be comprised of the sample selected for this examination and the number of defects acceptable shall be as specified in Table II.

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Defects

Overall uncleanness
Uneven weaving throughout piece
Selvage curls, folded, rolled, slack or loopy, torn continuously or intermittently over more than one half the roll

Table II - Acceptable number of defects

Lot size in yards	Sample size in rolls	Maximum number of defects applicable in sample
Up to 1300 1/	3	0
1301 to 3200	5	0
3201 to 8000	7	0
8001 to 22,000	10	0
22,001 to 110,000	15	1
110,001 and up	25	1

1/ If lot contains fewer than 3 rolls, each roll in the lot size shall be examined.

4.4.2.3 Examination for length.

4.4.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards (1.8m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of lengths marked on the ticket. The rolls examined shall be those selected for the examination of individual rolls.

4.4.2.4 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off-shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.4.2.5 Examination for non-conformance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for this defect. The lot shall be unacceptable if two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

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4.4.3 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be on the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5, expressed in terms of defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted container.
Content	Number of items per shipping container is more or less than required. Size shown on one or more items not as specified on shipping container. 1/

1/ For this defect, one item from each shipping container in sample shall be examined.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable, and as listed in Table III shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample unit for the test purposes shall be 3 continuous yards (2.7m) full width of the finished ribbon. The lot size shall be expressed in units of one yard. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table III - Test Methods

Characteristic	Requirement paragraph	Test Method
Material identification	3.3.1	1/
Denier	3.3.2	1/
Width	3.4	Visual 2/
Weight	3.4	5041
Warp ends:		
Ground	3.4	Visual 2/
Binder	3.4	Visual 2/
Selvage	3.4	Visual 2/
Filling		
Picks per inch	3.4	Visual 2/
Shrinkage		
Warp	3.4	5554
Filling	3.4	5554
Colorfastness		
Laundering	3.5.2	5614
Wet-Cleaning	3.5.2	5621
Light (40 hours)	3.5.2	5660 3/
Perspiration	3.5.2	5680
Crocking	3.5.2	AATCC No.8-88
Weave	3.6	Visual 2/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

3/ The calibration of the fadeometer and the definition of AATCC Fading Units shall be according to AATCC Test Method 16. Using the formula below, 16 to 22 AFU's are acceptable as 20 SFH's

$$\text{AATCC Fading Units} = \frac{(\Delta E^*)(20)}{1.7}$$

Where (ΔE^*) is the color difference in CIELAB* units of color difference.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Ribbon of one type and class only shall be wound on a paperboard or kraft flanged spool. The wound ribbon shall not extend beyond the edges of the flanges and the free end of the ribbon shall be secured with pressure sensitive tape. The spool shall contain 50 or 100 continuous yards ± 1/2 yard (45 or 90 continuous meters ± 45cm) as specified (see 6.2).

5.1.2 Level C. The ribbon shall be preserved to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Ribbon of the same type and class, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF, class weather-resistant, grade V15c of PPP-B-636. The inside of each shipping container shall be fitted with a taped box liner conforming to type CF, class weather-resistant, of PPP-B-636. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. The weight of the contents of the shipping container shall not exceed 65 pounds (29.25kg).

5.2.2 Level B. Ribbon of the same type and class, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF, class domestic, variety DW, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. The weight of the contents of each shipping container shall not exceed 65 pounds (29.25kg).

5.2.3 Level C. Ribbon of the same type and class, preserved as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

* 5.3.1 Containers. In addition to any special marking required by the contract or order, unit packs, intermediate or exterior containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

5.3.2 Spools. The outside surface of one flange of each spool shall be legibly marked with: stock number, name of the item, color, gross length, width and contract number.

* 5.4 MIL-STD-2073. When specified (see 6.2), Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The ribbon covered by this specification is intended for use in the fabrication of U.S. Coast Guard women's neckties.

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* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type and class required (see 1.2)
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)
d. When first article sample is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quality, testing and approval of the first article.
e. Put up lengths required (see 3.7 and 5.1.1).
f. Levels of preservation and packing required (see 5.1 and 5.2)
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.4).

6.3 Samples. For access to samples address the procuring activity issuing the invitation for bids (see 3.1).

* 6.5 Subject term (key word) listing.
Blue 3366
Fabric, necktie
Necktie, womens
Red 3805

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
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Preparing activity:
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Project 8305-N376

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-87017B(CG)	RIBBON, DECORATIVE-TRIMMING, POLYESTER, GROSGRAIN (CG)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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